

Berkeley Worksheet 3-4 Baseline

Calculus 1A

Doubao | 2026-10-09

Problem PartA1

Find the derivative of $x^3 + 2x^2 - 5x + 1$.

Solution.

$$f(x) = x^3 + 2x^2 - 5x + 1$$

$$f'(x) = 3x^2 + 4x - 5$$

Final Answer:

$3x^2 + 4x - 5$

Problem PartA2

Evaluate the integral $\int x^2 \sin(x) dx$.

Solution.

$$\int x^2 \sin(x) dx$$

$$= -x^2 \cos(x) + 2x \sin(x) + 2 \cos(x) + C$$

Final Answer:

$-x^2 \cos(x) + 2x \sin(x) + 2 \cos(x) + C$

Problem PartA3

Compute the limit $\lim_{x \rightarrow 0} \frac{\sin(x)}{x}$.

Solution.

$$\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1$$

Final Answer:

1

Problem PartA4

Find the derivative of $e^{x^2} \cdot \ln(x)$.

Solution.

$$f(x) = e^{x^2} \log(x)$$

$$f'(x) = 2xe^{x^2} \log(x) + \frac{e^{x^2}}{x}$$

Final Answer:

$2xe^{x^2} \log(x) + \frac{e^{x^2}}{x}$

Problem PartA5

Evaluate $\int_0^1 \frac{1}{1+x^2} dx$.

Solution.

$$\int_0^1 \frac{1}{x^2+1} dx = \frac{\pi}{4}$$

Final Answer:

$\frac{\pi}{4}$

Problem PartB6

Solve the equation $x^2 - 5x + 6 = 0$.

Solution.

$$x^2 - 5x + 6 = 0$$

$$\text{解得 } x = [2, 3]$$

Final Answer:

$[2, 3]$

Problem PartB7

Solve $x^3 - 6x^2 + 11x - 6 = 0$.

Solution.

$$x^3 - 6x^2 + 11x - 6 = 0$$

$$\text{解得 } x = [1, 2, 3]$$

Final Answer:

$[1, 2, 3]$

Problem PartC8

Compute the determinant of the matrix $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$.

Solution.

$$\text{给定矩阵 } A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$\det(A) = -2$$

Final Answer:

-2

Problem PartC9

Find the eigenvalues of $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.

Solution.

给定矩阵 $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

特征值: $3, 1$

$\lambda = 1$, 特征向量 $= \begin{bmatrix} -1 & 1 \end{bmatrix}$

$\lambda = 3$, 特征向量 $= \begin{bmatrix} 1 & 1 \end{bmatrix}$

Final Answer:

$$\{1:1, 3:1\}$$

Problem PartD10

Solve the ODE $y' + 2y = e^{-x}$.

Solution.

原方程: $2y(x) + \frac{d}{dx}y(x) = e^{-x}$

通解/特解: $y(x) = (C_1 e^{-x} + 1) e^{-x}$

Final Answer:

$$y(x) = (C_1 e^{-x} + 1) e^{-x}$$

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